# India Solar Panel Predictor

Stage 1 predicts temperature and sunlight for a state and 10-day period. Stage 2 turns those into panel voltage, power and energy. All models train live in your browser.

Predict | Models | Plan & data

Trained on 1976–2015, tested on 2016–2025 (unseen years). Default models: Climatology baseline → Linear Regression.

State: Maharashtra

Date: 23-12-2026

Panels in series (string): 10

Parallel strings: 2

Stage 1 model: Gradient Boosting

Stage 2 model: Gradient Boosting

Maharashtra · 23 Dec 2026 falls in period Dec 21–31 (period 36 of 36)

## Stage 1 · expected weather for this 10-day period

| | | |
|---|---|---|
| **28.7 °C** | **16.9 °C** | **5.24 kWh/m²/day** |
| Max temperature · range 26.9–28.9 °C | Min temperature · range 15.2–17.1 °C | Solar radiation · range 5.2–5.4 |

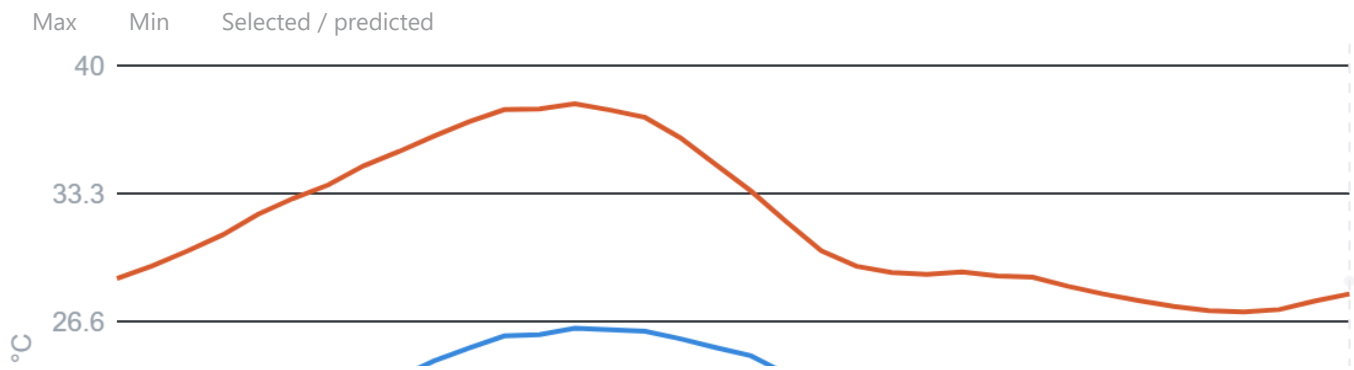
## Stage 2 · one panel (72-cell, 400 Wp)

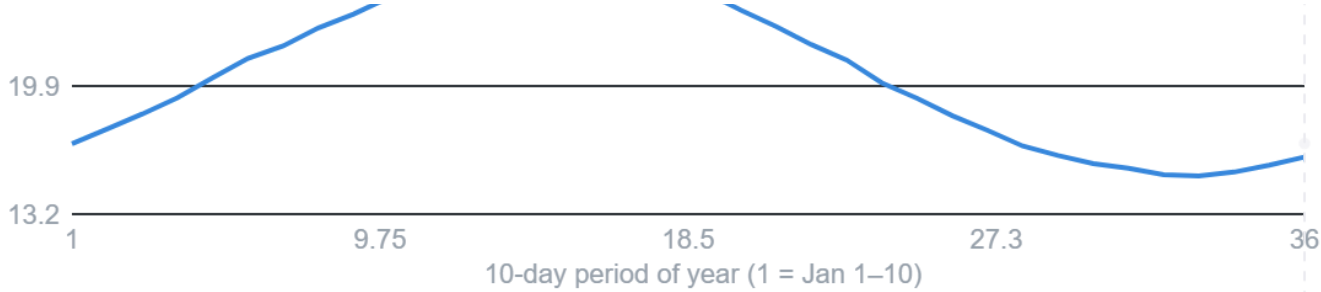
| | | | |
|---|---|---|---|
| **37.4 V** | **6.82 A** | **255 W** | **1.96 kWh/day** |
| Voltage at solar noon (Vmp) | Current at solar noon | Power at solar noon | Energy per day |

## Your array

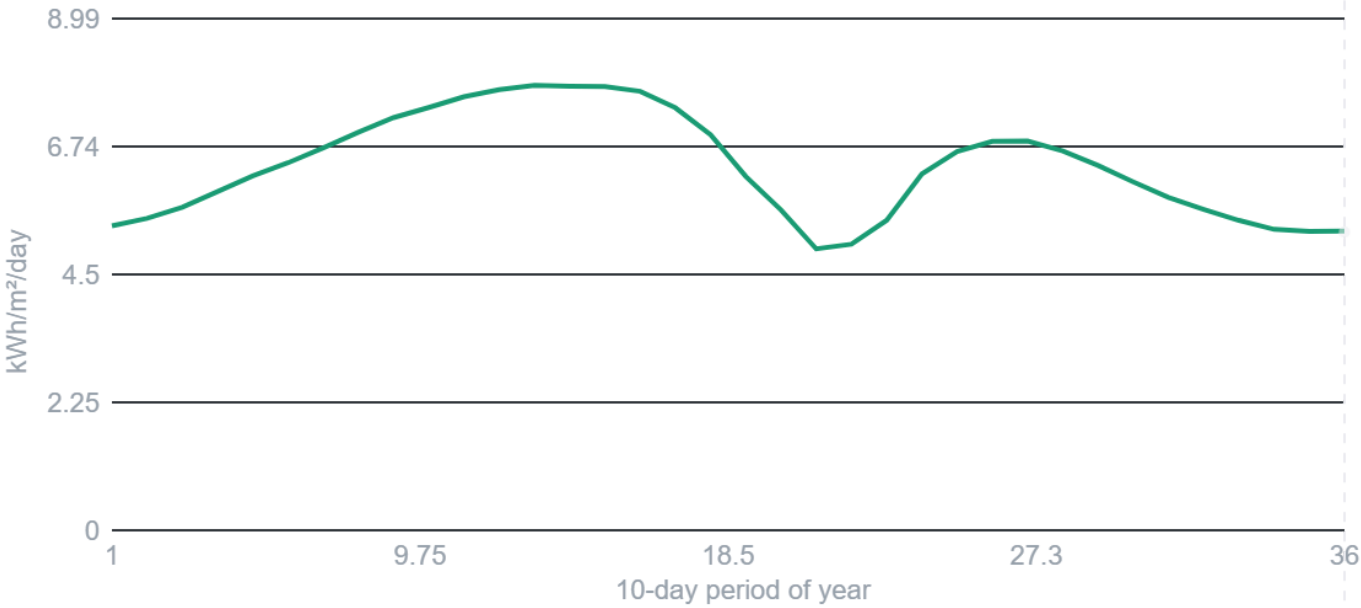
| | | | |
|---|---|---|---|
| **373.9 V** | **13.6 A** | **5.1 kW** | **33.6 kWh/day** |
| String voltage (10 in series) | Array current (2 strings) | Peak power (20 panels) | Energy, 14% losses · range 32.9–34.3 |

## Temperature by 10-day period (history mean)






## Solar radiation by 10-day period




These are typical conditions for the period (climate-style prediction), not a day-ahead weather forecast. Real days can be cloudier or clearer, so use the range shown.